

Final Model

```
In [9]: import joblib
import pandas as pd
import numpy as np

# Chargement du modèle final
pipeline = joblib.load("/Users/cordelia/Documents/PROJECT_12/model_final.
print("✅ Modèle chargé !")
```

✅ Modèle chargé !

```
In [10]: # Fonction de prédiction avec probabilités
FEATURES = ["diagonal", "height_left", "height_right", "margin_low", "mar

def predire_billets(data):
    """
    Prédit si un ou plusieurs billets sont vrais ou faux.
    data : dict (un billet) ou list de dicts (plusieurs billets)
    """
    df = pd.DataFrame([data] if isinstance(data, dict) else data)
    df = df[FEATURES]

    predictions = pipeline.predict(df)
    probas = pipeline.predict_proba(df)

    for i, (pred, proba) in enumerate(zip(predictions, probas)):
        statut = "✅ VRAI billet" if pred else "❌ FAUX billet"
        certitude = proba[1] if pred else proba[0]
        print(f"Billet {i+1} : {statut} – Certitude : {certitude:.2%}")
```

```
In [11]: # Test avec un billet
billet_test = {
    "diagonal": 171.81,
    "height_left": 104.86,
    "height_right": 104.95,
    "margin_low": 4.52,
    "margin_up": 2.89,
    "length": 112.83
}

predire_billets(billet_test)
```

Billet 1 : ❌ FAUX billet – Certitude : 52.95%

/Users/cordelia/Downloads/billets_production.csv

```
In [12]: df_test = pd.read_csv("/Users/cordelia/Downloads/billets_production.csv",
df_test.head()
```

Out [12]:

	diagonal	height_left	height_right	margin_low	margin_up	length	id
0	171.76	104.01	103.54	5.21	3.30	111.42	A_1
1	171.87	104.17	104.13	6.00	3.31	112.09	A_2
2	172.00	104.58	104.29	4.99	3.39	111.57	A_3
3	172.49	104.55	104.34	4.44	3.03	113.20	A_4
4	171.65	103.63	103.56	3.77	3.16	113.33	A_5

```
In [14]: def predire_depuis_csv(chemin_csv):
df_test = pd.read_csv(chemin_csv, sep=",")
ids = df_test["id"] if "id" in df_test.columns else range(1, len(df_test))
df_test = df_test[FEATURES]

predictions = pipeline.predict(df_test)
probas = pipeline.predict_proba(df_test)

for id_billet, pred, proba in zip(ids, predictions, probas):
    statut = "✅ VRAI billet" if pred else "❌ FAUX billet"
    certitude = proba[1] if pred else proba[0]
    print(f"{id_billet} : {statut} - Certitude : {certitude:.2%}")

predire_depuis_csv("/Users/cordelia/Downloads/billets_production.csv")
```

```
A_1 : ❌ FAUX billet - Certitude : 97.88%
A_2 : ❌ FAUX billet - Certitude : 99.54%
A_3 : ❌ FAUX billet - Certitude : 99.41%
A_4 : ✅ VRAI billet - Certitude : 90.05%
A_5 : ✅ VRAI billet - Certitude : 99.63%
```